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Nanoparticle fluids for an AI-assisted water alternating gas process

Abstract

A treatment fluid including an aqueous colloid containing a first surfactant and a plurality of nanoparticles encapsulated by a second surfactant. A method for preparing a treatment fluid including mixing a plurality of metallic oxide nanoparticles with a first surfactant to form an intermediate solution. A second surfactant is added to the intermediate solution to form nanoparticles encapsulated by the second surfactant. A method of extracting hydrocarbons from a well environment and storing carbon dioxide in the well environment including injecting a first amount of carbon dioxide and a first amount of a treatment fluid into a hydrocarbon reservoir via an injection well in a well environment. The treatment fluid treatment fluid including an aqueous colloid containing a first surfactant and a plurality of nanoparticles encapsulated by a second surfactant. Subsequently, determining a byproduct amount of the carbon dioxide extracted from the well environment.

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Background/Summary

BACKGROUND

(1) In the oil and gas industry, oil recovery methods may be used to produce or further produce hydrocarbons from a hydrocarbon reservoir within a formation. Each oil recovery method may be categorized as a primary, secondary, and/or tertiary recovery method. A water alternating gas (WAG) process has traditionally been categorized as a secondary or tertiary recovery method, which may be considered an enhanced oil recovery method. However, the WAG process may be performed at any time during the productive life of the hydrocarbon reservoir.

(2) WAG processes tend to cyclically alternate the injection of a miscible gas, such as carbon dioxide (CO₂), and water into the hydrocarbon reservoir to extract trapped hydrocarbons to the surface of the earth via a production well. Following the injection of miscible gas via an injection well, dissolution of the miscible gas and trapped hydrocarbons may occur to form a solution of mobilized hydrocarbons that reestablishes a hydrocarbon flow path within the hydrocarbon reservoir due to decreased viscosity of the solution relative to the trapped hydrocarbons. Following

the injection of water via the injection well, the water may flush the solution of mobilized hydrocarbons towards the production well to allow the solution of mobilized hydrocarbons to be extracted. In turn, as the water flushes the solution, some miscible gas may take the place of the hydrocarbons within the hydrocarbon reservoir and remain in the reservoir for storage.

(3) The environmental effects of CO.sub.2 on the atmosphere are of particular concern as CO.sub.2 is a “greenhouse gas” that may contribute to global warming. As such, sustainability efforts have become a recent societal goal to mitigate global warming. Altering the goals of the traditional WAG process to, at least in part, capture and store CO.sub.2 (i.e., CO.sub.2 sequestration) has emerged as a possible solution for reducing CO.sub.2 in the atmosphere to mitigate global warming.

SUMMARY

(4) This summary is provided to introduce a selection of concepts that are further described below in the detailed description. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.

(5) In one aspect, embodiments disclosed herein relate to a treatment fluid containing an aqueous colloid that is formed from a first surfactant and a plurality of nanoparticles encapsulated by a second surfactant.

(6) In another aspect, embodiments disclosed herein relate to a method for preparing a treatment fluid by mixing a plurality of metallic oxide nanoparticles with a first surfactant to form an intermediate solution and then adding a second surfactant to that intermediate solution, resulting in nanoparticles encapsulated by the second surfactant.

(7) In another aspect, embodiments disclosed herein relate to a method of extracting hydrocarbons from a well environment and storing carbon dioxide in the well environment. Initially, a first amount of carbon dioxide and a first amount of a treatment fluid is injected into a hydrocarbon reservoir through an injection well in the well environment. The treatment fluid contains an aqueous colloid that is formed from a first surfactant and a plurality of nanoparticles encapsulated by a second surfactant. Subsequently, a byproduct amount of carbon dioxide is extracted from the well environment.

(8) Other aspects and advantages of the claimed subject matter will be apparent from the following description and the appended claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 illustrates a process of preparing a treatment fluid in accordance with one or more embodiments.

(2) FIG. 2 illustrates a treatment system in accordance with one or more embodiments.

(3) FIG. 3 illustrates a treatment method in accordance with one or more embodiments.

(4) FIG. 4 illustrates a workflow in accordance with one or more embodiments.

(5) FIG. 5 illustrates a Markov decision process in accordance with one or more embodiments.

(6) FIG. 6 illustrates a neural network in accordance with one or more embodiments.

(7) FIG. 7 illustrates a computer system in accordance with one or more embodiments.

(8) FIG. 8 shows a flowchart in accordance with one or more embodiments.

(9) FIG. 9 shows a flowchart in accordance with one or more embodiments.

(10) FIG. 10 is a bar chart of interfacial tension measurements in accordance with one or more embodiments.

DETAILED DESCRIPTION

(11) In the following detailed description of embodiments of the disclosure, numerous specific details are set forth in order to provide a more thorough understanding of the disclosure. However,

it will be apparent to one of ordinary skill in the art that the disclosure may be practiced without these specific details. In other instances, well-known features have not been described in detail to avoid unnecessarily complicating the description.

(12) Throughout the application, ordinal numbers (e.g., first, second, third, etc.) may be used as an adjective for an element (i.e., any noun in the application). The use of ordinal numbers is not to imply or create any particular ordering of the elements nor to limit any element to being only a single element unless expressly disclosed, such as using the terms “before,” “after,” “single,” and other such terminology. Rather, the use of ordinal numbers is to distinguish between the elements. By way of an example, a first element is distinct from a second element, and the first element may encompass more than one element and succeed (or precede) the second element in an ordering of elements.

(13) It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a memory” includes reference to one or more of such memories.

(14) As used here and in the appended claims, the words “comprise,” “has,” and “include” and all grammatical variations thereof are each intended to have an open, non-limiting meaning that does not exclude additional elements or steps.

(15) “Optionally” and variations thereof means that the subsequently described event or circumstances may or may not occur. The description includes instances where the event or circumstance occurs and instances where it does not occur.

(16) When the word “approximately” or “about” and variations thereof are used, this term may mean that there can be a variance in value of up to +10%, of up to 5%, of up to 2%, of up to 1%, of up to 0.5%, of up to 0.1%, or up to 0.01%.

(17) The term “substantially” and variations thereof as used refer to a majority of, or mostly, as in at least about 50%, 60%, 70%, 80%, 90%, 95%, 96%, 97%, 98%, 99%, 99.5%, 99.9%, 99.99%, or at least about 99.999% or more.

(18) Ranges may be expressed as from about one particular value to about another particular value, inclusive. When such a range is expressed, it is to be understood that another embodiment is from the one particular value to the other particular value, along with all particular values and combinations thereof within the range.

(19) The phrases “artificial intelligence,” “machine learning,” “deep learning,” and “pattern recognition” are often interchanged and used synonymously. This variation of nomenclature tends to arise because the field of “extracting patterns and insights from data” was developed simultaneously and disjointedly among a number of classical arts like mathematics, statistics, and computer science. For consistency, the term “artificial intelligence” will be adopted herein and denoted “AI.” However, a person of ordinary skill in the art will recognize that the concepts and methods detailed hereafter are not limited by this choice of nomenclature. Further, in some fields, machine learning (ML) models may be considered a subset within AI models. As such, an ML model may more specifically describe the AI model or a portion of the AI model depending on the AI model selected.

(20) It is to be understood that one or more of the steps shown in the flowcharts may be omitted, repeated, and/or performed in a different order than the order shown.

(21) Accordingly, the scope disclosed herein should not be considered limited to the specific arrangement of steps shown in the flowcharts.

(22) Although multiple dependent claims are not introduced, it would be apparent to one of ordinary skill that the subject matter of the dependent claims of one or more embodiments may be combined with other dependent claims.

(23) In the following description of FIGS. 1-10, any component described regarding a figure, in various embodiments disclosed herein, may be equivalent to one or more like-named components described regarding any other figure. For brevity, descriptions of these components will not be

repeated regarding each figure. Thus, each and every embodiment of the components of each figure is incorporated by reference and assumed to be optionally present within every other figure having one or more like-named components.

(24) Additionally, in accordance with various embodiments disclosed herein, any description of the components of a figure is to be interpreted as an optional embodiment which may be implemented in addition to, in conjunction with, or in place of the embodiments described regarding a corresponding like-named component in any other figure.

(25) In one aspect, embodiments disclosed herein relate to the formulation of a treatment fluid containing encapsulated metal oxide nanoparticles and the process of preparing this treatment fluid for use. In other aspects, embodiments disclosed herein relate to a method to extract residual hydrocarbons from a formation using a water alternating gas process with the treatment fluid.

(26) Water alternating gas processes contain a miscible gas phase and an aqueous phase to extract residual hydrocarbons. It is standard practice to use water as the aqueous phase and carbon dioxide is used as the miscible gas phase. A water alternating gas process functions by alternating the injection of the miscible gas phase and the aqueous phase into a reservoir. The miscible gas phase dissolves into the trapped hydrocarbons to mobilize the hydrocarbons for extraction. The aqueous phase assists in flushing the mobilized hydrocarbons towards the extraction well. Using a metal oxide based nanofluid as a treatment fluid in place of water as the aqueous phase further reduces interfacial tension between the formation and the residual hydrocarbons, improving the yield of hydrocarbons in the extraction process. Simultaneously, the treatment fluid maximizes the solubility of carbon dioxide into the aqueous phase, improving the mobilization of the trapped hydrocarbons. The carbon dioxide may be at supercritical conditions when injected into the reservoir.

(27) In one or more embodiments, a treatment fluid contains nanoparticles, a first surfactant, a second surfactant, and brine. The nanoparticles may be first added to the first surfactant, followed by the second surfactant, with brine added in the last step. The first surfactant may include petroleum sulfonate and functions to provide a micelle around the nanoparticle, however, it is thermally unstable. The second surfactant may include cocamidopropyl hydroxysultaine. The role of the second surfactant is to encapsulate the micelles to form a stable capsule with an ionic strength of 75,000 ppm at a temperature of 100° C. for a year. The stable capsule may be in the form of a two layer membrane, where the initial micelle is a single layer and the second surfactant forms the second layer. The nanoparticles may be formed of one or more metal oxides. Suitable metal oxides include iron oxide, magnesium oxide, zinc oxide, aluminum oxide, and combinations thereof. The first surfactant may be in an amount ranging from about 5% to about 50% by weight of the treatment fluid. The second surfactant may be in an amount ranging from is about 5% to about 50% by weight of the treatment fluid.

(28) The brine includes water. The water may be distilled water, deionized water, tap water, fresh water from surface or subsurface sources, production water, formation water, natural and synthetic brines, brackish water, natural and synthetic sea water, black water, brown water, gray water, blue water, potable water, non-potable water, other waters, and combinations thereof, that are suitable for use in a wellbore environment. In one or more embodiments, the water used may naturally contain contaminants, such as salts, ions, minerals, organics, and combinations thereof, as long as the contaminants do not interfere with the tracer particle operations.

(29) FIG. 1 illustrates a process of preparing the treatment fluid. In block **100**, a plurality of metal oxide nanoparticles are mixed with a first surfactant in a vessel to form an intermediate solution. The vessel may contain an agitator to ensure adequate mixing. The nanoparticles may be in an amount ranging from about 0.02% to 0.08% by weight in about 5% to 50% by weight of the first surfactant in the treatment fluid. In block **110**, the intermediate solution mixes with a second surfactant in the same vessel to form encapsulated metal oxide nanoparticles. This process of preparing the treatment fluid may occur in a vessel located proximate the well environment. This

process may occur in a vessel at a different location and transported to the well environment either by a vehicle or by a process line. The present inventors have unexpectedly discovered that the sequential preparation of the treatment fluid as detailed above, results in encapsulated dispersed nanoparticles. "Dispersed nanoparticles" indicates that the encapsulated nanoparticles are evenly distributed throughout the treatment fluid, rather than forming a conglomeration of nanoparticles. (30) FIG. 2 illustrates a system for using the treatment fluid in a water alternating gas (WAG) process. Referring to FIG. 2, a fluid pumping system **212** contains an injection system **214** and a production system **216**. The injection system **214** includes an injection well **206** extending into a formation **204** under the earth's surface **222** and penetrating through a hydrocarbon reservoir **202**. The production system **216** includes a production well **208** extending into a formation **204** under the earth's surface **222** and penetrating through a hydrocarbon reservoir **202**. The injection system **214** is located at a distance across the hydrocarbon reservoir **202** from the production system **216**. The area of the hydrocarbon reservoir **202** containing the injection well **206** and the production well **208** and a plurality of WAG zones **250**, **248**, **242**, **238**, **240**, and **244** is considered the well environment. The plurality of WAG zones **250**, **248**, **242**, **238**, **240**, and **244** extends between the injection system **214** and the production system **216**. When injection well **206** and production well **208** are vertical wells, each WAG zone **250**, **248**, **242**, **238**, **240**, and **244** may form a layer, where the layers layer horizontally between the injection well **206** and **208**. Such a layer is termed herein a "horizontal layer." When an injection well and a production well are horizontal or lateral wells (not shown), each WAG zone may form a layer, where the layers layer vertically between the injection well and production well. Such a layer is termed herein "vertical layer." In an exemplary configuration WAG zones **250**, **248**, **242**, **238**, **240**, and **244** include aqueous zone **250**, first carbon dioxide zone **248**, accumulation zone **242**, second carbon dioxide zone **238**, miscible zone **240**, and oil zone **244**. Aqueous zone **250** contains a treatment fluid phase that includes the treatment fluid. The treatment fluid phase may be a drive phase that is initiated by a WAG cycle. First carbon dioxide zone **248** contains a carbon dioxide phase that includes carbon dioxide. The carbon dioxide in zone **248** may be supercritical at reservoir conditions. Accumulation zone **242** contains an accumulation phase. The accumulation phase may include fluid with accumulated nanoparticles and trapped carbon dioxide caused by a prior WAG cycle. Second carbon dioxide zone **238** contains a carbon dioxide phase that includes carbon dioxide. The carbon dioxide in zone **238** may be supercritical at reservoir conditions. Miscible zone **240** contains a miscible phase. The miscible phase may be caused by interaction of carbon dioxide and previous contact of residual oil with the treatment fluid. The previous contact may specifically be previous contact of residual oil with the nanoparticles in the treatment fluid. Oil zone **244** contains a mobilized oil phase. The mobilized oil phase may be caused by the WAG cycle. Zone **244** may also contain portions of the treatment fluid and the carbon dioxide as byproducts. The production system **216** is configured to extract hydrocarbons from zone **244**.

(31) It will be understood that through the present disclosure, where carbon dioxide is described, the carbon dioxide may be mixed with other miscible gases, such as, but not limited to, propane and butane, without departing from the scope of the disclosure.

(32) Referring still to FIG. 2, the production well **208** may use a production pump **252** to extract the mobilized hydrocarbons, byproduct treatment fluid, and byproduct carbon dioxide. The production well **208** may lead to a separation unit **261** through a production process line **260** containing a produced amount of hydrocarbons and byproduct amounts of the treatment fluid and carbon dioxide. The separation unit **261** may be in fluid communication with one or more hydrocarbon storage tanks **258**, one or more treatment fluid storage tanks **232**, and/or one or more carbon dioxide storage tanks **220**. Alternately or in combination, the treatment fluid and carbon dioxide or portions thereof may be routed directly back to the injection well **206** as a recycle stream without any storage unit. A combination of direct recycle streams and storage tanks may be used for the treatment fluid and/or the carbon dioxide. There may be a carbon dioxide process line **228**

containing a carbon dioxide pump **229** from the carbon dioxide storage tank **220** to the injection well **206**. There may be a treatment fluid process line **234** containing a treatment fluid pump **232** from the treatment fluid storage tank **232** to the injection well **206**. A tanker **224** may be used to provide carbon dioxide or the treatment fluid to the injection well **206**. Fluid pumping system **212** may be equipped to allow the tanker **224** to tie into either the carbon dioxide process line **228** or the treatment fluid process line **234**. The system contains a computer system **262** that communicates between the production well and the injection well to optimize the extraction of hydrocarbons and maximize a storage amount of carbon dioxide under the earth's surface **222**.

(33) Referring to FIG. 2, to initiate a WAG treatment process, the injection well **206** and the production well **208** are identified. The reservoir may be initially treated with standard hydrocarbon production processes. The reservoir may contain residual hydrocarbons, that is hydrocarbons produced by additional methods of extraction. Alternately, the reservoir may be untreated, containing larger volumes of hydrocarbons.

(34) Referring to FIG. 3, in block **310** of WAG process **300**, an injection amount a treatment fluid and an injection amount of carbon dioxide are injected in a repeatedly alternating pattern through the injection well into the reservoir. This alternating process may create different zones in the reservoir between the injection well and production well that lead to carbon dioxide mobilizing residual hydrocarbons for extraction from the production well. As the process progresses, the composition of each zone may vary. For example, if carbon dioxide is the component most recently injected, the closest zone in the reservoir to the injection well will be carbon dioxide, however, if the treatment fluid is the most recent component injected, the closest layer in the reservoir to the injection well will be the treatment fluid.

(35) Referring back to FIG. 2, this process schematic shows an embodiment where the most recently injected component is the treatment fluid. In FIG. 2, the treatment fluid is most recently injected and has formed aqueous zone **250** in the reservoir closest to the injection well. The first carbon dioxide zone **248** is adjacent to aqueous zone **250** and contains the carbon dioxide phase that was injected before the treatment fluid in aqueous zone **250** and is moving towards the production well **208**. The accumulation zone **242**, moving towards the production well, contains primarily the treatment fluid that was injected before the carbon dioxide layer **248**, with trapped carbon dioxide due to interactions between the treatment fluid and the carbon dioxide within the reservoir. The second carbon dioxide zone **238**, moving towards the production well, contains primarily carbon dioxide, similar to the first carbon dioxide zone **248**. The miscible zone **240**, moving towards the production well **208**, contains a miscible phase caused by the interaction between the carbon dioxide, the treatment fluid, and residual hydrocarbons. In the miscible zone **240**, the carbon dioxide is dissolving into the residual hydrocarbons and mobilizing the hydrocarbons. Within the miscible zone **240**, the treatment fluid is reducing interfacial tension between the hydrocarbons and the reservoir, further mobilizing the hydrocarbons. Within the miscible zone **240**, the treatment fluid is maximizing the solubility of carbon dioxide in the aqueous phase, further mobilizing the hydrocarbons as the treatment fluid flushes the hydrocarbons towards the production well. In this process, a portion of the injected carbon dioxide is stored in the hydrocarbon reservoir **202**. The oil zone **244** contains mobilized hydrocarbons along with a byproduct amount of the treatment fluid and carbon dioxide.

(36) Referring to FIG. 3, in block **320** of WAG process **300**, the water alternating gas process results in the extraction of a produced amount of hydrocarbons and a byproduct amount of the treatment fluid and carbon dioxide. In block **330** of WAG process **300**, the produced amount of hydrocarbons is separated from the byproduct amounts of the treatment fluid and carbon dioxide. The byproduct amounts of the treatment fluid and carbon dioxide are recycled to block **310** for injection into the system. Some of the carbon dioxide will remain in the reservoir as storage for carbon capture efforts. The quantity of carbon dioxide stored may depend on factors including geological related parameters, the solubility of carbon dioxide into the aqueous phase, and the mass

transfer of carbon dioxide into the oil phase. Geological related parameters include porosity, fracture network, and rock adsorption affinity. The WAG process **300** may maximize the affinity of carbon dioxide to adsorb or transfer into the rock, the aqueous phase, and the oil phase. The produced amount of hydrocarbons exits the system as the final product.

(37) Throughout WAG process **300**, an artificial intelligence controller scheme implemented in computer system **262** may adjust the injection flow rates and volumes of carbon dioxide and the treatment fluid to optimize the extraction of hydrocarbons and maximize the amount of carbon dioxide stored. The produced amount of hydrocarbons and the storage amount of carbon dioxide may be input into the artificial intelligence model, using a Markov decision process, to obtain an injection amount of hydrocarbons and an injection amount of carbon dioxide.

(38) The alternating pattern of WAG process **300** may include N cycles where N is an integer greater than or equal to one. For example, if N=2, the two cycles include a first cycle and second cycle. The count of cycles (i.e., first and second) are denoted n where n increments by one from one to N. In brief, for the traditional WAG process, each of the N cycles may include the injection of gas, such as carbon dioxide (CO.sub.2), into the hydrocarbon reservoir **202** followed by the injection of water into the hydrocarbon reservoir **202** further followed by the extraction of hydrocarbons from the hydrocarbon reservoir **202**.

(39) The injection system **214** may be configured to inject an nth set of injection amounts in turn into the hydrocarbon reservoir **202** via the injection well **206** for each of the N cycles. In the present WAG process, the nth set of injection amounts may include an nth amount of CO.sub.2 and an nth amount of treatment fluid. Further, the nth amount of the treatment fluid may be injected into the hydrocarbon reservoir **202** via the injection well **206** using an injection pump **218**. Note that each of the nth set of injection amounts may be colloquially referred to as “slugs” (e.g., a CO.sub.2 slug).

(40) Following injection of the nth amount of CO.sub.2 into the hydrocarbon reservoir **202** via the injection well **206** for the nth cycle **236**, the nth amount of CO.sub.2 floods the hydrocarbon reservoir **202** as illustrated in the first carbon dioxide zone **238** in FIG. 2. In turn, due to the miscibility of the CO.sub.2, the CO.sub.2 and the trapped hydrocarbons mutually dissolve to form a solution with reduced viscosity relative to the trapped hydrocarbons as illustrated in a miscible zone **240**. Note that miscibility may be defined as the ability of fluids to mix in all proportions. In turn, the interfacial tension (IFT) between the solution and the rock of the hydrocarbon reservoir **202** reduces to allow the solution to travel along now reestablished hydrocarbon flow path. Note that miscibility is a function of pressure. As such, the pressure of the hydrocarbon reservoir **202** during each cycle of the WAG process desirably ensures the hydrocarbon reservoir **202** is above a minimum miscibility pressure.

(41) Following the injection of the nth amount of the treatment fluid into the hydrocarbon reservoir **202**, the nth amount of the treatment fluid floods the hydrocarbon reservoir **202** as shown in an accumulation zone **142**. In turn, the nth amount of the treatment fluid may flush the solution in the miscible zone **240** into the oil zone **244**, which may contain mobilized hydrocarbons.

(42) If certain conditions within the hydrocarbon reservoir **202** are met, the solution within the oil zone **244** may be extracted to the surface of the earth **222** via the production well **208** as will be described in detail below. In turn, some of the CO.sub.2 within the solution may remain in the pores of the rock of the hydrocarbon reservoir **202** where the trapped hydrocarbons were.

(43) If certain conditions within the hydrocarbon reservoir **202** are not met, the solution within the miscible zone **240** and oil zone **244** may remain within the hydrocarbon reservoir **202** until the (n+1)th cycle **246** of the WAG process is performed. FIG. 2 illustrates the (n+1)th cycle **246** where an (n+1)th amount of CO.sub.2 is illustrated in a CO.sub.2 phase zone **248**, and an (n+1)th amount of the treatment fluid is illustrated in an aqueous zone **250**.

(44) Once the fluid within the oil zone **244** reaches the production well **208**, the production system **216** may be configured to extract the fluid from the hydrocarbon reservoir **202** using a production

pump **252** in fluid communication with the production well **208**. The amount of hydrocarbons for each of the N cycles may include an nth amount of hydrocarbons. The nth amount of hydrocarbons may be mixed with CO.sub.2 and/or the treatment fluid if CO.sub.2 and/or treatment fluid breakthrough has occurred.

(45) Meters (not shown) within the fluid pumping system **212** may measure and store WAG data associated with each of the N cycles of the WAG process. The WAG data may include the values associated with the nth amount of CO.sub.2 and the nth amount of the treatment fluid injected into the hydrocarbon reservoir **202**, the nth amount of stored carbon dioxide stored within the hydrocarbon reservoir **202** (i.e., CO.sub.2 sequestration), and the nth amount of produced hydrocarbons extracted from the hydrocarbon reservoir **202**. Hereinafter, the value associated with the nth amount of carbon dioxide injected into the hydrocarbon reservoir **202** may be referred to as an nth value of an injection amount of carbon dioxide. The value associated with the nth amount of the treatment fluid injected into the hydrocarbon reservoir **202** may be referred to as an nth value of an injection amount of the treatment fluid. The value associated with the nth amount of stored carbon dioxide stored within the hydrocarbon reservoir **202** may be referred to as an nth value of a stored amount of carbon dioxide. The value associated with the nth amount of produced hydrocarbons extracted from the hydrocarbon reservoir **202** may be referred to as an nth value of a produced amount of hydrocarbons. The term “amount” may refer to a volume, density, concentration, and/or rate. The WAG data may be transferred from the fluid pumping system **212** to a computer system **262** via a network as shown by the dashed lines in FIG. 2. In turn, the computer system **262** may be configured to receive, store, and/or process the WAG data. Further, data generated or obtained using the computer system **262** may be transferred to the fluid pumping system **212** via the network. Both the computer system **262** and network is described in detail relative to FIG. 7.

(46) The WAG process performed within the formation **204** as described in FIG. 2 may be modeled or simulated using a reservoir simulator. In these embodiments, the WAG data may be obtained from the reservoir simulator rather than measured and stored by meters within the fluid pumping system **212**.

(47) The reservoir simulator may mathematically model rock and the hydrocarbon reservoir **202** within the formation **204** as porous media. Further, the reservoir simulator may mathematically model fluid flow of CO.sub.2, the treatment fluid, and hydrocarbons within the formation **204** and the fluid pumping system **212** using principles of fluid dynamics, such as Navier-Stokes equations. In particular, the Navier-Stokes equations describe how the velocity, pressure, temperature, and density of a moving fluid are related. Because the N cycles of the WAG process occur over time, the reservoir simulator may further model the rock, hydrocarbon reservoir **202**, and fluid flow in the formation **204** over time for one or more WAG processes.

(48) Updating the reservoir simulator may be done using, but not limited to, finite difference methods, finite element methods, and finite volume methods. As such, the reservoir simulator may be stored on a memory of a computer system **262**, which, again, is described relative to FIG. 7. If the reservoir simulator is rendered as an image, the image may illustrate one or more injection wells **206** and one or more production wells **208** within the formation **204**. However, a person of ordinary skill in the art will appreciate that the reservoir simulator may not be rendered as an image and only output values of interest. Hereinafter, the reservoir simulator is assumed to model one injection well **206** and one production well **208** to mimic the formation **204** illustrated in FIG. 2. However, a person of ordinary skill in the art will appreciate that the reservoir simulator may model more than one injection well **206** and/or more than one production well **206** without departing from the scope of the disclosure.

(49) In one or more embodiments, the reservoir simulator is used to generate one or more outputs based on inputs that are fed into the reservoir simulator. These input/output values are used to form M training pairs <input(s), output(s)>, where M is an integer greater than or equal to one. For

example, if $M=2$, the two training pairs include a first training pair and a second training pair. The count of training pairs (i.e., first and second) are denoted m , where m increments by one from one to M . For example, if $m=1$, an m th training pair describes a first training pair. The count m may describe a current cycle number like n .

(50) Each of the M training pairs or the m th training pair includes an m th set of injection values **400** (input(s)) and an m th set of output values **408** as shown in FIG. 4. The m th set of injection values **400** includes an m th value of an injection amount of CO.sub.2 **402** that may be injected into the modeled hydrocarbon reservoir within the reservoir simulator **404**. The m th set of injection values **400** further includes an m th value of an injection amount of the treatment fluid **406** that may be injected into the modeled hydrocarbon reservoir within the reservoir simulator **404**. Each of the m th set of injection values **400** may be assigned randomly, based on historical WAG data, based on a probability distribution, and/or based on the current states of the formation **204** or the modeled formation, among other methods.

(51) The m th set of injection values **400** may be input into the reservoir simulator **404** to generate the m th set of output values **408**. The m th set of output values **408** may include an m th value of a stored amount of CO.sub.2 **410** stored within the modeled hydrocarbon reservoir within the reservoir simulator **404**. The m th set of output values **408** may further include an m th value of a produced amount of hydrocarbons **412** extracted from the modeled hydrocarbon reservoir within the reservoir simulator **404**.

(52) The m th set of output values **408** may be directly obtained from the reservoir simulator **404**. In other embodiments, the m th set of output values **408** may be sampled from an m th set of output probability distributions that may be directly obtained from the reservoir simulator **404**. In these embodiments, the sampling method may be based on random sampling, Monte Carlo sampling, bootstrapping, or any other sampling method.

(53) The process illustrated in FIG. 4 may be repeated to generate each of the M training pairs. In practice, tens to thousands of training pairs may be generated. The M training pairs may be used to train, at least in part, an artificial intelligence (AI) model.

(54) The AI model may include a Markov decision process (MDP). FIG. 5 illustrates the MDP **500** in accordance with one or more embodiments. The MDP **500** may rely on an agent **502** to interact with the formation **204** to determine actions **504** based on the current states **506** of the formation **204**. Here, the formation **204a** may be the physical formation **204** illustrated in FIG. 2 or the modeled formation within the reservoir simulator **404** as described relative to FIG. 4. The agent **502** uses its policy **508** to determine the actions **504** based, at least in part, on the current states **506** of the formation **204a**.

(55) For example, assume the current states **506** of the formation **204a** are denoted $s_{\text{sub}.n}$, which may come from a set of states S . The agent **502** may receive the current states $s_{\text{sub}.n}$ **506** from the formation **204a**. The agent **502** may then use its policy **508** to determine the next actions $a_{\text{sub}.n+1}$ **504**, which may be determined from a set of actions A , (once n increases by one as shown in block **510**) that may be taken within the formation **204a**. This process may continue for the N cycles of the WAG process. In other embodiments, this process may continue indefinitely where $N \rightarrow \infty$.

(56) The agent **502** may learn what actions **504** produce desirable states and what actions **504** produce undesirable states by observing the consequences of the actions **504** (i.e., the updated states). This process may be referred to as reinforcement learning. In these embodiments, the agent **502** may learn and alter its behavior to use its policy T_t **508** to determine actions **504** that produce desirable states by mapping from states **506** to a probability distribution p over actions **504** where:

(57) $\pi: S \rightarrow p(A = a_n | S)$. Equation(1)

(58) The agent **502** may learn and alter its behavior based on a rewards system or rewards function R . When actions **504** produce desirable states, a positive reward may be assigned as the reward

r.sub.n **512** and passed to the agent **502**. When actions **504** produce undesirable states, a negative reward or penalty may be assigned as the reward r.sub.n **512** and passed to the agent **502**. For each “trajectory” or “rollout” of the policy **508**, the agent **502** learns based on the previous states s.sub.n-1, the previous actions a.sub.n-1, the current states s.sub.n **506**, and the reward r.sub.n **512** as illustrated in FIG. 5 to improve its policy **508**. The goal of the agent **502** is to determine a policy **508** that maximizes the expected reward **512** where:

(59)
$$\pi^* = \operatorname{argmax} \mathbb{E}[R]. \quad \text{Equation(2)}$$

(60) In the context of this disclosure, desirable states may include storing an increased amount of CO.sub.2 in the hydrocarbon reservoir **202** or the modeled hydrocarbon reservoir for the current cycle n relative to the previous cycle n-1 and extracting an increased amount of hydrocarbons from the hydrocarbon reservoir **202** or the modeled hydrocarbon reservoir for the current cycle n relative to the previous cycle n-1. In other words, additional CO.sub.2 is stored and additional hydrocarbons are extracted after each cycle N.

(61) The mapping performed by the policy **508** may be done by a neural network. In other embodiments, the AI model may be a neural network. As such, an introduction to neural networks is provided herein. While the term “neural network” is used hereinafter, a person of ordinary skill in the art will appreciate that the neural network may be a specialized neural network, such as a recurrent neural network (RNN) or long short-term memory (LSTM), without departing from the scope of the disclosure. It is emphasized that the following discussion of the neural network is a basic summary that should not be considered limiting as many variations of neural networks, RNNs, and LSTMs exist. Therefore, one of ordinary skill in the art will recognize that any variation of a neural network, RNN, or LSTM may be employed without departing from the scope of the disclosure.

(62) FIG. 6 illustrates a neural network **600** in accordance with one or more embodiments. At a high level, a neural network **600** may be graphically depicted as being composed of nodes **602** and edges **604**. The nodes **602** may be grouped to form layers. FIG. 6 displays four layers **608**, **610**, **612**, **614** of nodes **602** where the nodes **602** are grouped into columns. However, each group need not be as shown in FIG. 6. The edges **604** connect the nodes **602** to other nodes **602**. Edges **604** may connect, or not connect, to any node(s) **602** regardless of which layer **608**, **610**, **612**, **614** the node(s) **602** is in. That is, the nodes **602** may be sparsely and residually connected. For example, in an RNN, nodes **602** in the output layer **614** may be connected by edges **604** to nodes **602** in the input layer **608** (though not shown in FIG. 6). As such, an RNN may be considered an autoregressive process.

(63) A neural network **600** will have at least two layers, where the first layer **608** is the “input layer” and the last layer **614** is the “output layer.” Any intermediate layer **610**, **612** is usually described as a “hidden layer.” A neural network **600** may have zero or more hidden layers **610**, **612**. A neural network **600** with at least one hidden layer **610**, **612** may be described as a “deep” neural network or “deep learning method.” If the AI model includes an MDP **500** that uses reinforcement learning and the policy **508** includes a deep neural network, the AI model may be considered to use deep reinforcement learning.

(64) In general, a neural network **600** may have more than one node **602** in the output layer **614**. In these cases, the neural network **600** may be referred to as a “multi-target” or “multi-output” network.

(65) Nodes **602** and edges **604** carry associations. Namely, every edge **604** is associated with a numerical value. The edge numerical values, or even the edges **604** themselves, are often referred to as “weights” or “parameters.” While training a neural network **600**, a process that will be described below, numerical values are assigned to each edge **604**. Additionally, every node **602** is associated with a numerical value and may also be associated with an activation function. Activation functions are not limited to any functional class but traditionally follow the form:

(66) $A = f(. \text{Math}.i \in (\text{incoming}) [(\text{nodevalue})_i (\text{edgevalue})_i])$, Equation(3)

where i is an index that spans the set of “incoming” nodes **602** and edges **604** and f is a user-defined function. Incoming nodes **602** are those that, when viewed as a graph (as in FIG. 6), have directed arrows that point to the node **602** where the numerical value is being computed. Some functions f may include the linear function $f(x)=x$, sigmoid function

(67) $f(x) = \frac{1}{1+e^{-x}}$,

and rectified linear unit function $f(x)=\max(0,x)$, however, many additional functions are commonly employed. Every node **602** in a neural network **600** may have a different associated activation function. Often, as a shorthand, activation functions are described by the function f by which it is composed. That is, an activation function composed of a linear function f may simply be referred to as a linear activation function without undue ambiguity.

(68) When the neural network **600** receives an input, the input is propagated through the neural network **600** according to the activation functions and incoming node values and edge values to compute a value for each node **602**. That is, the numerical value for each node **602** may change for each received input while the edge values remain unchanged. Occasionally, nodes **602** are assigned fixed numerical values, such as the value of one. These fixed nodes **606** are not affected by the input or altered according to edge values and activation functions. Fixed nodes **606** are often referred to as “biases” or “bias nodes” as displayed in FIG. 6 with a dashed circle.

(69) In some implementations, the neural network **600** may contain specialized layers, such as a normalization layer, pooling layer, or additional connection procedures, like concatenation. One of ordinary skill in the art will appreciate that these alterations do not exceed the scope of the disclosure.

(70) The number of layers in a neural network **600**, choice of activation functions, inclusion of batch normalization layers, and regularization strength, among others, may be described as “hyperparameters” that are associated to the neural network **600**. It is noted that in the context of AI, the regularization of the neural network **600** refers to a penalty applied to the loss function of the neural network **600**. The selection of hyperparameters associated to a neural network **600** is commonly referred to as selecting the neural network “architecture.”

(71) Once a neural network **600** and associated hyperparameters have been selected, the neural network **600** may be trained. To do so, the M training pairs as previously described may be provided to the neural network **600**. In general, each of the M training pairs includes an input (i.e., the m th set of output values **408**) and an associated target output (i.e., the m th set of injection values **400**). Each associated target output represents the “ground truth,” or the otherwise desired output upon processing the input. During training, the neural network **600** processes at least one input from the M training pairs to produce at least one output. Each neural network output is then compared to the associated target output from the M training pairs.

(72) Returning to the neural network **600** in FIG. 6, the neural network **600** may be trained by first assigning initial values to the edges **604**. These values may be assigned randomly, according to a prescribed distribution, manually, or by some other assignment mechanism. Once edge values have been initialized, the neural network **600** may act as a function such that it may receive an input from the M training pairs and produce an output. At least one input is propagated through the neural network **600** to produce an output.

(73) The comparison of the neural network output to the associated target output from the M training pairs is typically performed by a “loss function.” Other names for this comparison function include an “error function,” “misfit function,” and “cost function.” Many types of loss functions are available, such as the log-likelihood function or cross-entropy loss function. However, the general characteristic of a loss function is that the loss function provides a numerical evaluation of the similarity between the neural network output and the associated target output from the M training pairs. The loss function may also be constructed to impose additional constraints on the values

assumed by the edges **604**. For example, a penalty term, which may be physics-based, or a regularization term may be added. Generally, the goal of a training procedure is to alter the edge values to promote similarity between the neural network output and associated target output for most, if not all, of the M training pairs. Thus, the loss function is used to guide changes made to the edge values. This process is typically referred to as “backpropagation.”

(74) While a full review of the backpropagation process exceeds the scope of this disclosure, a brief summary is provided. Backpropagation consists of computing the gradient of the loss function over the edge values. The gradient indicates the direction of change in the edge values that results in the greatest change to the loss function. Because the gradient is local to the current edge values, the edge values are typically updated by a “step” in the direction indicated by the gradient. The step size is often referred to as the “learning rate” and need not remain fixed during the training process. Additionally, the step size and direction may be informed by previous edge values or previously computed gradients. Such methods for determining the step direction are usually referred to as “momentum” based methods.

(75) Once the edge values of the neural network **600** have been updated through the backpropagation process, the neural network **600** will likely produce different outputs than it did previously. Thus, the procedure of propagating at least one input from the M training pairs through the neural network **600**, comparing the neural network output with the associated target output from the M training pairs with a loss function, computing the gradient of the loss function with respect to the edge values, and updating the edge values with a step guided by the gradient is repeated until a termination criterion is reached. Common termination criteria include, but are not limited to, reaching a fixed number of edge updates (otherwise known as an iteration counter), reaching a diminishing learning rate, noting no appreciable change in the loss function between iterations, or reaching a specified performance metric as evaluated on the M training pairs or separate hold-out training pairs (denoted “validation pairs”). Once the termination criterion is satisfied, the edge values are no longer altered and the neural network **600** is said to be “trained.”

(76) Following the selection of the neural network **600** and associated hyperparameters, training, and validation, the neural network **600** may be deployed for use. As such, the neural network **600** may be used to obtain the nth set of injection values (i.e., the actions a.sub.n **504**) from the nth set of output values (i.e., the states s.sub.n **506**).

(77) The neural network **600** may be retrained using additional WAG data obtained from the reservoir simulator **404** and/or measured from the formation **204**. In these embodiments, retraining may be considered to use transfer learning.

(78) In these embodiments, the nth set of output values may be input into the AI model. The nth set of output values may be the states **506** of the formation **204** described in FIG. 2 or the modeled formation within the reservoir simulator **404** described in FIG. 4. The (n+1)th set of injection values may then be obtained from the AI model. Next, the (n+1)th set of injection values may be either input into the reservoir simulator **404** or used to determine an nth amount of CO.sub.2 and an nth amount of the treatment fluid to be injected in turn into the hydrocarbon reservoir **202** within the formation **204**. In turn, an (n+1)th set of output values may be obtained from the reservoir simulator **404** or an (n+1)th amount of hydrocarbons may be extracted from the hydrocarbon reservoir **202** within the formation **204** while an (n+1)th value of the stored amount of carbon dioxide is determined from the formation **204**. The AI model may then be rewarded or penalized based on the nth set of output values and the (n+1)th set of output values. The AI model may then be retrained using the (n+1)th set of injection values and the (n+1)th set of output values. This process may be repeated for any number of N cycles.

(79) Turning to FIG. 7, FIG. 7 illustrates a generic computer system **262** in accordance with one or more embodiments. Hereinafter, FIG. 7 is used to describe one computer system **262** configured to communicate with the fluid pumping system **212** via the network **700** and store both the reservoir simulator **404** and the AI model **702** within a memory **704** each of which may be used by the

computer system **262**. However, a person of ordinary skill in the art will appreciate that more than one computer system **262** may be used without departing from the scope of the disclosure.

(80) For example, in other embodiments, a different computer system **262**, which may or may not be configured to communicate with the fluid pumping system **212** via the network **700**, may be configured to store the reservoir simulator **404**, AI model **702**, or both. Further, one computer system **262** may be configured to train and use the AI model **702**. In other embodiments, one computer system **262** may be configured to train the AI model **702** while another computer system **262** may be configured to use the AI model **702**.

(81) Each component of the computer system **262** illustrated in FIG. 7 will now be described in detail.

(82) The computer system **262** may be configured to provide computational functionalities associated with described AI models **702**, reservoir simulators **404**, WAG processes and other algorithms, methods, functions, processes, flows, and procedures as described in this disclosure, according to one or more embodiments. The illustrated computer system **262** is intended to encompass any computing device such as a server, desktop computer, laptop/notebook computer, wireless data port, smart phone, personal data assistant (PDA), tablet computing device, one or more processors **706** within these devices, or any other suitable processing device, including both physical or virtual instances (or both) of the computing device. Additionally, the computer system **262** may include an input device, such as a keypad, keyboard, touch screen, or other device that can accept user information, and an output device that conveys information associated with the operation of the computer system **262**, including digital data, visual, or audio information (or a combination of information), or a graphical user interface (GUI).

(83) The computer system **262** can serve in a role as a client, network component, a server, a database or other persistency, or any other component (or a combination of roles) for performing the subject matter described in the instant disclosure. The illustrated computer system **262** is communicably coupled with the network **700**. In some implementations, one or more components of the computer system **262** may be configured to operate within environments, including cloud-computing-based, local, global, or other environment (or a combination of environments).

(84) At a high level, the computer system **262** is an electronic computing device operable to receive, transmit, process, store, or manage data and information associated with the described subject matter. According to some implementations, the computer system **262** may also include or be communicably coupled with an application server, e-mail server, web server, caching server, streaming data server, business intelligence (BI) server, or other server (or a combination of servers).

(85) The computer system **262** can receive requests over the network **700** from a client application (for example, executing on another computer system **262**) and responding to the received requests by processing the said requests in an appropriate software application. In addition, requests may also be sent to the computer system **262** from internal users (for example, from a command console or by other appropriate access method), external or third-parties, other automated applications, as well as any other appropriate entities, individuals, systems, or computers.

(86) Each of the components of the computer system **262** can communicate using a system bus **708**. In some implementations, any or all of the components of the computer system **262**, both hardware or software (or a combination of hardware and software), may interface with each other or the interface **710** (or a combination of both) over the system bus **708** using an application programming interface (API) **712** or a service layer **714** (or a combination of the API **712** and service layer **714**). The API **712** may include specifications for routines, data structures, and object classes. The API **712** may be either computer-language independent or dependent and refer to a complete interface, a single function, or even a set of APIs. The service layer **714** provides software services to the computer system **262** or other components (whether or not illustrated) that are communicably coupled to the computer system **262**. The functionality of the computer system **262**

may be accessible for all service consumers using this service layer. Software services, such as those provided by the service layer **714**, provide reusable, defined business functionalities through a defined interface. For example, the interface may be software written in JAVA, C++, or other suitable language providing data in extensible markup language (XML) format or another suitable format. While illustrated as an integrated component of the computer system **262**, alternative implementations may illustrate the API **712** or the service layer **714** as stand-alone components in relation to other components of the computer system **262** or other components (whether or not illustrated) that are communicably coupled to the computer system **262**. Moreover, any or all parts of the API **712** or the service layer **714** may be implemented as child or sub-modules of another software module, enterprise application, or hardware module without departing from the scope of this disclosure.

(87) The computer system **262** includes an interface **710**. Although illustrated as a single interface **710** in FIG. 7, two or more interfaces **710** may be used according to particular needs, desires, or particular implementations of the computer system **262**. The interface **710** is used by the computer system **262** for communicating with other systems in a distributed environment that are connected to the network **700**. Generally, the interface **710** includes logic encoded in software or hardware (or a combination of software and hardware) and operable to communicate with the network **700**. More specifically, the interface **710** may include software supporting one or more communication protocols associated with communications such that the network **700** or interface's hardware is operable to communicate physical signals within and outside of the illustrated computer system **262**.

(88) The computer system **262** includes at least one computer processor **706**. Although illustrated as a single computer processor **706** in FIG. 7, two or more processors may be used according to particular needs, desires, or particular implementations of the computer system **262**. Generally, the computer processor **706** executes instructions and manipulates data to perform the operations of the computer system **262** and any algorithms, methods, functions, processes, flows, and procedures as described in the instant disclosure.

(89) The computer system **262** also includes a memory **704** that holds data for the computer system **262** or other components (or a combination of both) that can be connected to the network **700**. For example, memory **704** can be a database storing data consistent with this disclosure. Although illustrated as a single memory **704** in FIG. 7, two or more memories may be used according to particular needs, desires, or particular implementations of the computer system **262** and the described functionality. While memory **704** is illustrated as an integral component of the computer system **262**, in alternative implementations, memory **704** can be external to the computer system **262**.

(90) The application **716** is an algorithmic software engine providing functionality according to particular needs, desires, or particular implementations of the computer system **262**, particularly with respect to functionality described in this disclosure. For example, application **716** can serve as one or more components, modules, applications, etc. Further, although illustrated as a single application **716**, the application **716** may be implemented as multiple applications **716** on the computer system **262**. In addition, although illustrated as integral to the computer system **262**, in alternative implementations, the application **716** can be external to the computer system **262**.

(91) There may be any number of computer systems **262** associated with, or external to, a computer system **262** containing a computer system **262**, wherein each computer system **262** communicates over the network **700**. Further, the term “client,” “user,” and other appropriate terminology may be used interchangeably as appropriate without departing from the scope of this disclosure. Moreover, this disclosure contemplates that many users may use one computer system **262**, or that one user may use multiple computer systems **262**.

(92) FIG. 8 describes a method of training an AI model **702** in accordance with one or more embodiments. Turning to FIG. 8, M training pairs are obtained. Recall that M is an integer greater

than or equal to one. Each of the M training pairs may be generically denoted an m th training pair, where m is a count of the training pairs by one from one to M . Further, each of the M training pairs (i.e., the m th training pair) includes an m th set of injection values **400** and an m th set of output values **408**.

(93) In block **805**, the m th set of injection values **400** are input into the reservoir simulator **404**. As previously described relative to FIGS. **2** and **4**, the reservoir simulator **404** may mathematically model one or more WAG processes being performed within a formation **204**. The reservoir simulator **404** may mathematically model the rock and the hydrocarbon reservoir **202** within the formation **204** as porous media. The reservoir simulator **404** may mathematically model fluid flow of CO₂, the treatment fluid, and the hydrocarbons using principles of fluid dynamics. The m th set of injection values **400** input into the reservoir simulator **404** includes the m th value of the injection amount of CO₂ **402** and the m th value of the injection amount of the treatment fluid **406**. Each value may correspond to a volume, density, concentration, and/or rate.

(94) In block **810**, the m th set of output values **408** are generated from the reservoir simulator **404**. The m th set of output values **408** includes the m th value of the stored amount of CO₂ that may be stored within the modeled hydrocarbon reservoir of the reservoir simulator **404** and the m th value of the produced amount of hydrocarbons **412** that may be extracted from the modeled hydrocarbon reservoir of the reservoir simulator **404**. In some embodiments, the m th set of output values **408** are directly generated from the reservoir simulator **404**. In other embodiments, an m th set of output probability distributions are generated from the reservoir simulator **404** and the m th set of output values **408** are sampled from the m th set of output probability distributions. Sampling methods may include, but are not limited to, random sampling, Monte Carlo sampling, and bootstrapping. In practice, tens to thousands of training pairs may be generated.

(95) In block **815**, the AI model **702** is trained using the M training pairs. The AI model **702** may include the neural network **600** that is trained. In other embodiments, the AI model **702** may include a MDP **500** that includes a policy **508** that includes the neural network **600** that is trained. As previously described relative to FIG. **6**, the process of backpropagation may use the M training pairs to train the neural network **600** within the AI model **702**. Following training, the AI model **702** may be deployed for use to obtain an $(n+1)$ th set of injection values from an n th set of output values. Here, n is a count by one from one to N by where N is an integer greater than or equal to one.

(96) The AI model **702** may be retrained one or more times using additional data obtained from the AI model **702**, the reservoir simulator **404**, the formation **204**, or a combination thereof. In these embodiments, the AI model **702** may use reinforcement learning in the form of a rewards system as previously described.

(97) FIG. **9** describes a method of extracting hydrocarbons from the formation **204** and storing CO₂ in the formation **204** using the trained AI model **702** in accordance with one or more embodiments. In one or more embodiments, the method uses the treatment fluid. In other embodiments, the method uses a fluid, which may be water. The method described in FIG. **9** may be performed for each of N cycles where the current cycle is denoted the n th cycle. Recall that N is an integer greater than or equal to one and n is a count from one to N by integer.

(98) Here, the trained AI model **702** may be considered a controller that uses reinforcement learning to facilitate meeting the contradictory goals of mitigating the release of greenhouse gases into the atmosphere while increasing the extraction of the hydrocarbons from the hydrocarbon reservoir **202** for each cycle of the non-traditional WAG process.

(99) In block **905**, the n th set of output values is input into the trained AI model **702**. The n th set of output values includes an n th value of the stored amount of CO₂ within the formation **204** and an n th value of a produced amount of hydrocarbons extracted from the formation **204**. The n th set of output values may be the current states $s_{sub.n}$ **506** of the formation **204** as described in FIG. **5**. In other embodiments, the n th set of output values may be assigned randomly, based on historical

WAG data, or based on data obtained from the reservoir simulator **404**. Here, the trained AI model **702** has been previously trained one or more times.

(100) In block **910**, the (n+1)th set of injection values is obtained from the trained AI model **702**. The (n+1)th set of injection values includes an (n+1)th value of an injection amount of CO.sub.2 and an (n+1)th value of an injection amount of the treatment fluid. The (n+1)th set of injection values may be the current actions a.sub.n **504** to be implemented within the formation **204** as described in FIGS. **2** and **4**.

(101) In block **915**, an (n+1)th amount of CO.sub.2 and an (n+1)th amount of the treatment fluid are injected into the hydrocarbon reservoir **202** in turn based on the (n+1)th set of injection values. Injection may be performed using the injection system **214** within the fluid pumping system **212** as described in FIG. **2**. The injection system **214** may inject the (n+1)th amount of CO.sub.2 and the (n+1)th amount of the treatment fluid into the hydrocarbon reservoir **202** via the injection well **206** within the formation **204**.

(102) In block **920**, an (n+1)th set of output values is determined from the formation. To do so, an (n+1)th amount of hydrocarbons is extracted from the hydrocarbon reservoir **202**. Extraction may be performed using the production system **216** within the fluid pumping system **212** as described in FIG. **2**. The production system **216** may extract the (n+1)th amount of hydrocarbons from the hydrocarbon reservoir **202** via the production well **208** within the formation **204**. Further, an (n+1)th value of the produced amount of hydrocarbons is determined from the (n+1)th amount of hydrocarbons. Further still, an (n+1)th value of the stored amount of CO.sub.2 is determined within the hydrocarbon reservoir **202**. The (n+1)th value of the produced amount of hydrocarbons and the (n+1)th value of the stored amount of CO.sub.2 may be measured and stored using meters within the fluid pumping system **212**.

(103) In block **925**, the trained AI model **702** is rewarded or penalized based, at least in part, on the nth set of output values and the (n+1)th set of output values. The trained AI model **702** may be rewarded or assigned a positive reward if the current states **506** of the formation **204** are desirable. Desirable states may occur when an increased amount of CO.sub.2 is stored in the hydrocarbon reservoir **202** for the current cycle (n+1) relative to the previous cycle n and an increased amount of hydrocarbons is extracted from the hydrocarbon reservoir **202** for the current cycle (n+1) relative to the previous cycle n. In other words, additional CO.sub.2 is stored and additional hydrocarbons are extracted after each cycle N. Undesirable states may occur when only one or neither of the desirable states occur.

(104) Embodiments of the present disclosure may provide at least one of the following advantages. Using the water alternating gas process with the treatment fluid allows for more effective extraction of residual hydrocarbons because of the impact this treatment fluid has on the reduction in interfacial tension and increase in solubility of carbon dioxide in the aqueous phase (the treatment fluid). These properties of the treatment fluid allow an improved extraction of hydrocarbons. While optimizing extraction of residual hydrocarbons, this process provides a means for simultaneous sub-surface carbon dioxide storage to reduce environmental carbon dioxide emissions. The artificial intelligence model further improves the yield of hydrocarbons by optimizing the injection amounts of carbon dioxide and the treatment fluid, while simultaneously maximizing sub-surface carbon dioxide storage.

EXAMPLES

(105) Hereinafter, examples are provided that demonstrate the encapsulation, increased stability, and reduced IFT as described in the Detailed Description section.

(106) The treatment fluid demonstrates high stability under expected reservoir temperatures and pressures due to the composition and its encapsulation. The metal oxide nanoparticles are particularly beneficial over other nanoparticles, such as silica-based nanoparticles, because of their smaller size and higher stability in reservoir conditions. Iron oxide nanoparticles are exemplary of metal oxide nanoparticles. Further, the paramagnetic characteristics of the iron oxide nanoparticles

allow for mapping during the injection phase, which is useful in situations with a shallow formation, as the magnetic nanoparticles can be coupled with electromagnetic mapping to characterize the formation and enhance production from that zone.

(107) In order to evaluate the stability of the treatment fluid, treatment fluid was tested in a laboratory setting at temperatures above 70° C. and above salinities of 30,000 ppm. The treatment fluid demonstrated high colloidal stability at 90° C. in the presence of hot brine for nearly a year. Dynamic light scattering analysis and interfacial tension measurements provided data to evaluate the effectiveness and stability of the treatment fluid. The encapsulation was verified using Cryo-transmission electron microscopy (TEM) which showed empty capsules after following a simple mixing procedure, and successfully encapsulated nanoparticles after following the procedure outlined in FIG. 1. TEM images demonstrate the structure of the nanoparticles. The images show that a portion of the encapsulated nanoparticles contain multi-layer membranes while others are single layer, potentially due to the coalescence of some capsules together. The images detect encapsulated nanoparticles over 5 nm.

(108) Referring to FIG. 10, the interfacial tension (IFT) was quantified in the presence of the treatment fluid for four concentrations of iron-oxide nanoparticles by % weight. Table 1 shows the composition of the four fluids used. In the table, the metal oxide nanoparticles are referred to as SPIONS, or superparamagnetic iron oxide nanoparticles.

(109) TABLE-US-00001 TABLE 1 Compositions of nanofluids used for testing Petronate 5 nm wt. % of SPIONS in 5 wt. % Sample ID HL/L (g) SPIONS (mL) Petronate HL/L solution 0 0.5 0 0 NF 1 0.5 0.3 0.02 NF 2 0.5 0.6 0.04 NF 3 0.5 0.9 0.08

(110) The interfacial tension is shown to decrease by approximately 50% for concentrations of nanoparticles in the treatment fluid ranging from about 0.02% to 0.08% by weight. The resulting stability data and demonstrated reduction in interfacial tension in the presence of the treatment fluid indicated suitability of this formulation for enhanced hydrocarbon recovery applications. The interfacial tension was evaluated at varying concentrations of nanoparticles in the treatment fluid, including 0.02%, 0.04%, and 0.08% and showed reductions in the interfacial tension ranging between 45% and 55%. In the tests, the aqueous phase was a seawater and crude oil was utilized. Measurements were done at 95° C. after the samples were incubated in the oven in the presence of the crude oil for over 24 hours until the interfacial tension values were equilibrated. The interfacial tension was then measured using a Kruss spinning drop instrument.

(111) In addition to quantifiable measurements, two sets of samples were created for each nanofluid concentration assessed above. In the first set of samples, the encapsulation was conducted with the second surfactant and the nanoparticles. In the second set of samples, the encapsulation was conducted with the first surfactant and the nanoparticles. The results showed visible differences at time increments of 24 hours, 48 hours, and 72 hours. Notably, the first set of samples showed instability, as the samples began to form an agglomeration within the fluid once 72 hours had elapsed. The second set of samples showed even dispersion consistently between the start time and 72 hours.

(112) The encapsulation, stability in brine, and reduced IFT of the treatment fluid demonstrated that the treatment fluid may be suitable for use in WAG processes to enhance hydrocarbon recovery.

(113) Although only a few example embodiments have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the example embodiments without materially departing from this invention. Accordingly, all such modifications are intended to be included within the scope of this disclosure as defined in the following claims.

Claims

1. A treatment fluid, comprising: an aqueous colloid comprising: a brine; 0.02 to 0.08% by weight of a nanoparticle; 5% to 50% by weight of a first surfactant in the form of a micelle around the

nanoparticle, wherein the first surfactant comprises petroleum sulfonate; and a second surfactant encapsulating the micelle around the nanoparticle.

2. The treatment fluid of claim 1, wherein the second surfactant comprises cocamidopropyl hydroxysultaine.

3. The treatment fluid of claim 1, wherein the nanoparticle comprises a metal oxide.

4. The treatment fluid of claim 3, wherein the metal oxide is selected from the group consisting of iron oxide, magnesium oxide, zinc oxide, aluminum oxide, and combinations thereof.

5. The treatment fluid of claim 1, wherein the treatment fluid comprises the second surfactant in an amount ranging from about 5% to 50% by weight of the treatment fluid.

6. The treatment fluid of claim 1, wherein the treatment fluid is stable for a year or more under conditions of a temperature of 70 degrees Celsius ($^{\circ}$ C.) or more and a salinity of 30,000 parts per million (ppm) or more.

7. A method for preparing a treatment fluid, comprising: mixing a metal oxide nanoparticle with a first surfactant to form an intermediate solution containing a metal oxide nanoparticle capsule in the form of a micelle around the nanoparticle, wherein the first surfactant comprises petroleum sulfonate; and adding a second surfactant to the intermediate solution, wherein a sequential addition of the first surfactant and the second surfactant encapsulates the metal oxide nanoparticle capsule to form the treatment fluid; wherein the treatment fluid comprises 0.02% to 0.08% by weight of the metal oxide nanoparticles and 5% to 50% by weight of the first surfactant.

8. The method of claim 7, wherein the second surfactant comprises cocamidopropyl hydroxysultaine.

9. The method of claim 7, wherein the nanoparticle comprises a metal oxide.

10. A method of extracting hydrocarbons from a well environment and storing carbon dioxide in the well environment comprising: injecting, via an injection well in the well environment, a first amount of carbon dioxide and a first amount of a treatment fluid in turn into a hydrocarbon reservoir; extracting, via a production well in the well environment, a produced amount of the hydrocarbons from the hydrocarbon reservoir; and determining a byproduct amount of the carbon dioxide extracted from the well environment; wherein the treatment fluid comprises an aqueous colloid comprising a nanoparticle and a first surfactant wherein the first surfactant is in the form of a micelle around the nanoparticle; and a second surfactant encapsulating the micelle around the nanoparticle; wherein the treatment fluid comprises 0.02% to 0.08% by weight of the metal oxide nanoparticles and 5% to 50% by weight of the first surfactant.

11. The method of claim 10, further comprising: inputting a value of the byproduct amount of the carbon dioxide and a value of the produced amount of hydrocarbons into a trained artificial intelligence model; and obtaining a value of a second amount of carbon dioxide and a value of a second amount of the treatment fluid from the trained artificial intelligence model.

12. The method of claim 11, wherein the trained artificial intelligence model comprises a Markov decision process.

13. The method of claim 10, further comprising separating the produced amount of hydrocarbons as a final product from a byproduct amount of the treatment fluid and the byproduct amount of the carbon dioxide.

14. The method of claim 10, further comprising recycling the byproduct amount of the treatment fluid and the byproduct amount of the carbon dioxide for reinjecting.
